

Question description

Hassan enjoys jumping from one building to the next. However, he merely jumps to the next higher building and stops when there are none accessible. The amount of stamina necessary for a voyage is equal to the xor of all the heights Hassan leaps till he comes to a halt.

If heights are [1 2 4], and he starts from 1, goes to 2 stamina required is $1 \oplus 2 = 3$, then from 2 to 3. Stamina for the entire journey is $1 \oplus 2 \oplus 4 = 7$. Find the maximum stamina required if can start his journey from any building.

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq \text{Height} \leq 10^9$$

Input

First line: N , no of buildings.

Second line: N integers, defining heights of buildings.

Output

Single Integer is the maximum stamina required for any journey.

Explanation:

8

Question No:8

1 2 3 8 6 4 7 9

Considering the input given by you, 1st starting point is 1. From this building, Hassan will try to find the next higher building that is building 2 here and so on. Likewise he will complete all pass.

Pass 1: 1-> 2->3->8-> 9, XOR = 1

Pass 2: 2->3->8-> 9, XOR = 0

Pass 3: 3->8-> 9, XOR = 2

Pass 4: 8-> 9, XOR = 1

Pass 5: 6->7->9 , XOR = 8

Pass 6: 4->7->9 , XOR = 10

Pass 7: 7-> 9 XOR = 14

Pass 8: 9 XOR = 9

So, the highest XOR value is 14

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
1 2 3 8 6

EXPECTED OUTPUT

11

Test Case 2

INPUT (STDIN)

8
1 2 3 8 6 4 7 9

EXPECTED OUTPUT

14

Code:

```
1  #include <stdio.h>
2
3  #define MAX 100000
4
5  int main()
6  {
7      int N;
8      int height[MAX];
9      int stack[MAX];
10     int next[MAX];
11     int top = -1;
12
13     scanf("%d", &N);
14
15     for (int i = 0; i < N; i++)
16     {
17         scanf("%d", &height[i]);
18         next[i] = -1;
19     }
20
21     /*
22      Find the next greater element for every building.
23      stack stores indices.
24     */
25     for (int i = 0; i < N; i++)
26     {
27         while (top != -1 && height[i] > height[stack[top]])
28         {
29             next[stack[top]] = i;
30             top--;
31         }
32         stack[++top] = i;
33     }
34
35     /*
36      For every possible starting building,
```

```
38     follow the next-greater path and calculate XOR.
39     */
40     int maxXor = 0;
41
42     for (int i = 0; i < N; i++)
43     {
44         int current = i;
45         int xorValue = 0;
46
47         while (current != -1)
48         {
49             xorValue = xorValue ^ height[current];
50             current = next[current];
51         }
52
53         if (xorValue > maxXor)
54         {
55             maxXor = xorValue;
56         }
57     }
58
59     printf("%d", maxXor);
60
61     return 0;
62 }
```

Output:

Test Case-1:

```
5
1 2 3 8 6
11
-----
Process exited after 14.82 seconds with return value 0
Press any key to continue . . . |
```

Test Case-2:

```
8
1 2 3 8 6 4 7 9
14
-----
Process exited after 3.549 seconds with return value 0
Press any key to continue . . . |
```

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